PROJECT REPORT

A Fail-Safe Monocular Perception Pipeline for Collision and Overtaking Decisions in Unstructured Traffic

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All results reported in this document are measured from executed runs. Sections carrying unmeasured values are marked explicitly.

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Abstract

Driver-assistance systems developed against Western and East Asian benchmarks degrade on unstructured roads, where lane markings are intermittent and the vehicle population includes categories those benchmarks never recorded. This report documents a six-stage monocular camera perception pipeline that performs object detection, multi-object tracking, lane and drivable-corridor estimation, metric depth recovery, time-to-collision computation, and an overtaking safety verdict in a single forward pass per frame.

Two design commitments distinguish the system. First, every stage substitutes a weaker method rather than aborting when a model or weight file is unavailable — a property verified in deployment when the lane network failed to initialise and the pipeline nonetheless completed all 3,604 frames of the evaluation sequence. Second, quantities informing safety decisions are expressed in physical units: closing speed is differentiated from metric depth rather than image-plane motion, and lane curvature is projected onto the ground plane so the resulting radius can be compared against road-geometry design standards instead of an uncalibrated pixel threshold.

On a merged validation split of 1,496 images containing 8,104 annotated instances the detector attains 95.4 % mAP@0.5 and 79.9 % mAP@0.5:0.95, with infrequent classes matching or exceeding the aggregate figure. The complete pipeline sustains 23.1 frames per second on an NVIDIA H100 accelerator. A geometric plausibility filter removes detections whose implied physical size is inconsistent with their measured depth, rejecting 13.8 % of size-checked candidates. The report states the system's untrained capabilities explicitly and traces each to a data deficiency rather than an implementation defect.

CHAPTER 1

Introduction

1.1 Background

Perception stacks for driver assistance are overwhelmingly developed against a small family of benchmarks. KITTI [1] dominates object detection and CULane [2] dominates lane estimation. Both were recorded where traffic is lane-disciplined, markings are continuous, and the set of vehicle types is small and homogeneous.

Systems tuned on this material transfer poorly to roads that violate those premises. Varma *et al.* [3] documented the discrepancy quantitatively, observing that segmentation models trained on structured-road corpora score substantially lower on Indian road scenes than on their native distribution.

1.2 Problem Statement

The degradation is not uniform. It manifests as three distinct failure modes, each requiring a different remedy — a distinction that matters, because treating all three as one problem leads to the wrong fix being applied.

1.2.1 Category omission

A detector whose label space lacks a class cannot abstain from it; it assigns the nearest available label. A three-wheeled passenger vehicle becomes a car; a hand-cart becomes miscellaneous clutter. Downstream logic then reasons about an object whose dimensions and kinematics it has fundamentally misjudged. This is more hazardous than a missed detection, because the system acts with confidence on a wrong premise.

1.2.2 An absent estimation target

Lane-line detectors regress a curve through painted markings. Where the paint is faded, absent, or disregarded by traffic, the quantity being estimated does not exist in the scene. Retraining the lane model cannot repair this: the deficiency lies in the scene, not the estimator.

1.2.3 Spurious activation

Under distribution shift a detector's confidence becomes a less faithful indicator of correctness. Boxes appear on painted advertisements, shopfront imagery, and specular reflections — objects with no physical extent at all.

1.3 Objectives

1. Build a single-pass perception pipeline producing two actionable driver outputs: an imminent-collision alert and an overtaking verdict.
2. Ensure each stage degrades to a weaker method rather than terminating the pipeline when a component is unavailable.

3. Express safety-relevant quantities in physical units rather than image units, so thresholds refer to external standards.
4. Suppress spurious detections using geometric consistency rather than confidence thresholding alone.
5. Evaluate end to end and report untrained capabilities explicitly.

1.4 Scope and Non-Goals

The system is camera-only. Active range sensing would yield more reliable distances, and the monocular choice should be read as a deployability trade-off rather than a claim of superiority. The system is a perception and advisory layer; it issues no control commands.

CHAPTER 2

Literature Review

2.1 The Domain-Shift Problem

The India Driving Dataset [3] established that benchmarks recorded in countries with well-developed road infrastructure do not transfer directly to unstructured traffic. It introduced an expanded label hierarchy and, notably, explicit *fallback* categories for road regions and vehicles that structured taxonomies cannot express. The authors' framing of the problem is the foundation of this project:

> Datasets like KITTI, Cityscapes, Argoverse, and nuScenes are captured in developed countries where infrastructure is well-developed and road activity is structured. . . results obtained from these datasets are often not directly applicable in unstructured road situations prevalent in large parts of the world.
>
> — *Varma et al., WACV 2019*

Subsequent corpora reinforced the finding at larger scale. DriveIndia [4] contributes 66,986 annotated images spanning 24 traffic categories collected over more than 3,400 km of driving, reporting a best mAP@0.5 of 78.7 % among YOLO-family baselines. The UVH-26 release [5] contributes 26,646 traffic-camera images with 14 India-specific vehicle classes and reports improvements of up to 31.5 % in mAP@0.5:0.95 for detectors fine-tuned on this material relative to COCO-trained initialisations. Together these establish both that the gap is substantial and that supervised adaptation closes a large fraction of it.

2.2 Component Architectures

2.2.1 Object detection

Single-stage detection descends from the reformulation of detection as direct regression introduced by Redmon *et al.* [6]. This work adopts the YOLO11 implementation [7], whose anchor-free head and decoupled classification and regression branches provide a favourable accuracy-to-latency ratio at the input resolution the latency budget permits.

2.2.2 Lane estimation

CLRNet [8] detects lane candidates from high-level semantic features and refines them against lower-level detail, introducing a line-based intersection-over-union objective that regresses each lane as a unit rather than as independent points. It is reported to perform particularly well on curved and nocturnal subsets — the conditions of greatest interest here. Ultra-Fast Lane Detection v2 [9], which formulates lane localisation as hybrid-anchor ordinal classification, is retained as a lower-latency alternative.

2.2.3 Multi-object tracking

ByteTrack [10] observes that low-confidence detections frequently correspond to genuine but occluded objects, and matches them against unmatched tracks in a second association stage. Continuity of identity is a precondition for the temporal differencing the collision estimate depends on.

Table 2.1: Findings from the literature and the design decisions they produced

Finding	Design consequence
Western and East Asian benchmarks do not transfer to unstructured roads	Label space extended from 11 to 15 categories; converters built for three Indian corpora
COCO-trained detectors miss three-wheelers and light commercial vehicles entirely	The observed failure was attributed to data rather than implementation, redirecting effort accordingly
Sequential fine-tuning risks forgetting the original categories	Corpora merged and trained jointly rather than in sequence
Fine-tuning on Indian data yields up to +31.5 points mAP@0.5:0.95	Adopted as the measurable target for the adaptation experiment
Detectors hallucinate objects under distribution shift	Geometric plausibility filter added, rather than raising the confidence threshold alone
Apparent size and estimated depth must agree for a detection to be physical	That agreement test is the filter's mechanism
Unstructured roads require drivable-region segmentation, not lane lines	Segmentation fallback that activates automatically

2.2.4 Monocular depth

Depth Anything V2 [11] provides a metric-calibrated outdoor variant returning absolute distances, removing the scale ambiguity that would otherwise prevent a monocular system from reporting time-to-collision in seconds.

2.2.5 Low-light enhancement

The zero-reference curve estimation of Li *et al.* [12] requires no paired day–night supervision, which suits a project without access to registered night-time ground truth.

2.3 Consistency and Reliability

The plausibility filter developed in this work follows the observation from the monocular 3D detection literature [13] that detectors are susceptible to inconsistency between an object's apparent size and its estimated depth, and that enforcing agreement between the two is an effective corrective. The principle is applied here at the level of detection admission rather than 3D box regression.

2.4 Synthesis: What the Survey Determined

Table 2.1 records how each finding translated into a concrete design decision. This mapping is the substantive output of the review.

2.5 Research Gaps

1. Benchmark corpora encode structured roads; transfer to unstructured traffic is poor and quantified only for isolated tasks, not for a full decision-producing pipeline.
2. Lane-line estimation is undefined where markings are absent; the ground truth itself does not exist.

3. Spurious detections under distribution shift are typically addressed by confidence thresholding, which trades recall for precision rather than testing physical plausibility.
4. Curvature reported in pixels has no fixed physical meaning, yet is used to inform safety decisions.
5. Published systems address these problems in isolation; none combines detection, tracking, lanes, depth, and two decision layers under an explicit degradation contract.

CHAPTER 3

System Architecture

3.1 Overview

Six stages execute in sequence on every frame. Each consumes the frame together with the outputs of the stages above it; no second pass over the sequence is required. Figure 3.1 presents the flow.

3.2 Stage Descriptions

Stage 0 — Scene lighting. Classifies illumination from the frame's own intensity statistics and enhances only when the scene is genuinely dark, so daytime sequences incur no cost.

Stage 1 — Object detection. Detects road users at 640 px input resolution.

Stage 1b — Plausibility filter. Removes physically impossible candidates, as described in Section 4.2.

Stage 2 — Lane and corridor. Estimates lane geometry, or the drivable corridor where markings are unusable.

Stage 3 — Tracking. Associates detections across frames and maintains smoothed velocity.

Stage 4 — Depth and collision. Recovers metric depth and derives collision estimates.

Stage 5 — Overtaking verdict. Fuses the preceding outputs into a permitted / not-permitted decision.

3.3 Degradation Contract

Every optional component is constructed inside a guarded initialiser. Failure to load leaves the corresponding capability disabled and is reported at start-up, but does not propagate. Consumers of an unavailable output fall back to a weaker but valid estimate.

This is not a defensive convenience. It was exercised in deployment when the lane network failed to initialise owing to a library incompatibility, and the pipeline completed the full 3,604-frame sequence using the drivable-corridor substitute.

INPUT		monocular forward camera — video file, image sequence, or live stream
0	**Scene lighting** Zero-DCE++, CLAHE fallback → *enhanced frame, lighting state*	
1	**Object detection** YOLO11x at 640 px → *boxes, class, confidence*	
1b	**Plausibility filter** depth–size consistency, pinhole model → *physically admissible detections*	
2	**Lane and corridor** CLRNet ResNet-101; surface segmentation fallback → *lane polylines, ego-lane boundaries*	
3	**Multi-object tracking** two-stage IoU association, smoothed velocity → *persistent identifiers, velocity*	
4	**Metric depth and TTC** Depth Anything V2, metric outdoor → *distance, closing speed, TTC, alert tier*	
5	**Overtaking verdict** five-rule conjunctive fusion, no learned component → *permitted / not permitted, with cause*	
OUTPUT		annotated video and a per-frame decision record

Figure 3.1: The six-stage pipeline. Stages run in order on every frame, each consuming the outputs above it. Every stage substitutes a weaker method rather than aborting when its model is unavailable.

CHAPTER 4

Methodology

4.1 Collision Estimation from Metric Depth

An object's growth in the image plane is not equivalent to its approach in the world; scale change confounds relative motion with lateral displacement and camera pitch. Closing speed is therefore derived from depth rather than pixel kinematics.

For a track i, distance d_i is the median depth over the lower half of the bounding box — the region corresponding to the object's contact with the road surface. The upper half is excluded because glazing and reflective bodywork corrupt monocular depth there. The median is preferred to the mean for resistance to outlying pixels.

Maintaining a bounded history of d_i, the closing speed over n frames at frame rate f is

$$v_i = \frac{d_i^{(t-n)} - d_i^{(t)}}{n/f}, \tag{4.1}$$

positive when approaching, and the time-to-collision is

$$\tau_i = \frac{d_i^{(t)}}{v_i}, \qquad v_i > v_{\min}. \tag{4.2}$$

The threshold $v_{\min}$ suppresses alerts on objects whose apparent approach is attributable to depth noise. Alerts are emitted at two severities and only for objects within the ego corridor, so oncoming and roadside traffic does not generate spurious warnings. Distance history is nevertheless maintained for *all* tracks, including those outside the corridor, because the overtaking stage requires the closing speed of oncoming vehicles.

4.2 Plausibility Filtering

Under the pinhole model, an object of image width w_{px} at depth d observed with focal length f_{px} subtends a physical width

$$W = \frac{w_{\text{px}}\, d}{f_{\text{px}}}. \tag{4.3}$$

For each class an admissible interval $[W_{\min}, W_{\max}]$ is defined, and candidates whose implied W falls outside it after a multiplicative tolerance are discarded. A detection fired on printed imagery or a reflection has no consistent depth signature at its apparent scale and is removed, whereas a genuine vehicle satisfies the relation. Classes of genuinely unbounded extent are exempted rather than forced into an arbitrary interval.

Where the camera is uncalibrated, f_{px} must be estimated from an assumed field of view. Because such an estimate can err in either direction, the tolerance is *widened* in the uncalibrated case. This asymmetry is deliberate: discarding a real vehicle is more consequential than admitting a spurious one, since spurious detections remain subject to the ego-corridor test before they can raise an alert. Supplying a calibration restores the narrow interval automatically.

4.3 Curvature in Physical Units

Curvature computed on image-plane polylines has no fixed physical interpretation: it varies with sensor resolution, focal length, and mounting geometry, so the same road yields different values on different vehicles.

A plane-induced homography [14] projects lane points onto the ground plane. A second-order polynomial $x = az^2 + bz + c$ is fitted in ground coordinates, and

$$\kappa = \frac{|2a|}{\left(1 + (2az_f + b)^2\right)^{3/2}} \tag{4.4}$$

is evaluated at the furthest visible distance z_f, giving a radius $R = 1/\kappa$ in metres. This admits comparison against minimum horizontal curve radii from road-geometry design practice, so a verdict that a curve is too tight for a safe overtaking manoeuvre rests on an external standard rather than a tuned constant.

4.4 Overtaking Verdict

The verdict is a conjunction of five conditions, evaluated without a learned component:

1. marking legality,
2. curve radius,
3. visibility,
4. oncoming traffic within the manoeuvre window,
5. clear distance ahead of the lead vehicle.

All five must hold. Any input that is missing or uncertain resolves to *not permitted*. The module is conservative by construction, which is the correct property for a manoeuvre that places the vehicle in an opposing lane.

CHAPTER 5

Data Pipeline and Experimental Setup

5.1 Preprocessing

Source annotations were converted to a common normalised format and remapped onto a unified label space so that corpora with differing taxonomies could be combined. Boxes smaller than two pixels in either dimension were discarded and all coordinates clipped to the image. The checkpoint evaluated here was trained on 5,985 images with 1,496 held out for validation.

Augmentation comprised mosaic composition, MixUp, copy–paste, HSV value jitter, rotation to $\pm 5°$, translation, scaling, and horizontal flipping.

5.2 Unified Label Space

Fifteen categories, extended beyond a structured-road taxonomy with *autorickshaw*, *animal*, *rider*, and an open-world *vehicle fallback* class following the argument of [3]:

0	car	4	pedestrian	8	traffic light	12	animal
1	truck	5	cyclist	9	traffic sign	13	rider
2	bus	6	motorcycle	10	miscellaneous	14	vehicle fallback
3	van	7	tram				

As Chapter 7 discusses, the merge used for this checkpoint exercised only a subset of these.

5.3 Training Configuration

AdamW with an initial learning rate of 5×10^{-4} under a cosine schedule with five warm-up epochs; batch size 16 at 640×640; mixed precision; early stopping with patience 50. All experiments ran on a single NVIDIA H100 80 GB accelerator on a PBS-scheduled cluster.

5.4 Verification Procedure

Because training occupies the accelerator for extended periods, the decision logic — collision arithmetic, all five overtaking conditions, lane coordinate mapping, plausibility filtering, and the degradation contract — is exercised against synthetic inputs by a 62-check suite requiring no accelerator, trained weights, or dataset. It completes in seconds and is executed before any job is queued.

The suite detected several defects, including two that had already reached the cluster: a coordinate-space error in which lane predictions expressed in the lane model's native resolution were applied directly to frames of a different resolution, and an over-broad exception boundary that allowed an optional component's failure to terminate the pipeline.

CHAPTER 6

Results

6.1 Detection Accuracy

Table 6.1 reports per-class performance on the held-out split. Aggregate accuracy is 95.4 % mAP@0.5 and 79.9 % mAP@0.5:0.95, at 95.4 % precision and 92.5 % recall over 8,104 instances.

Table 6.1: Per-class detection performance (1,496 validation images, 8,104 instances)

Class	Instances	P (%)	R (%)	mAP@0.5	mAP@0.5:0.95
Car	5,798	95.7	96.6	97.9	87.6
Pedestrian	884	95.0	81.3	89.0	55.9
Van	587	94.9	97.3	97.5	86.0
Cyclist	313	94.3	88.2	92.7	71.0
Truck	219	97.5	98.6	99.4	91.5
Miscellaneous	194	95.1	92.3	94.3	80.8
Tram	109	95.3	93.1	96.8	86.7
All	**8,104**	**95.4**	**92.5**	**95.4**	**79.9**

Two observations merit attention.

Accuracy on infrequent categories does not collapse. Truck (219 instances) and tram (109) match or exceed the aggregate figure despite being roughly twenty-five times rarer than car, indicating that the balancing and augmentation strategy suppressed the long-tail degradation typically seen on such distributions.

Pedestrian localisation is the weakest column at 55.9 % under the strict metric, while precision remains at 95.0 %. Pedestrians are found reliably but bounded less tightly than vehicles, which is attributable to greater pose variability and more frequent partial occlusion.

Residual confusion is concentrated between visually adjacent vehicle categories rather than distributed arbitrarily, and the background row indicates that missed detections rather than cross-class errors dominate the remaining loss.

6.2 Effect of Training Schedule and Capacity

Table 6.2 compares three configurations. Extending the medium model from 50 to 500 epochs yields a larger improvement (+2.5 points mAP@0.5, +8.4 points at the strict metric) than the subsequent increase in capacity (+1.2 and +2.2 points). Under a constrained accelerator budget, schedule length was the more productive expenditure.

6.3 End-to-End Pipeline

Table 6.3 summarises a complete run over a 3,604-frame road sequence. All six stages sustain 23.1 frames per second at a mean latency of 43 ms.

An initial measurement of the same configuration returned 0.9 frames per second. The cause was place-

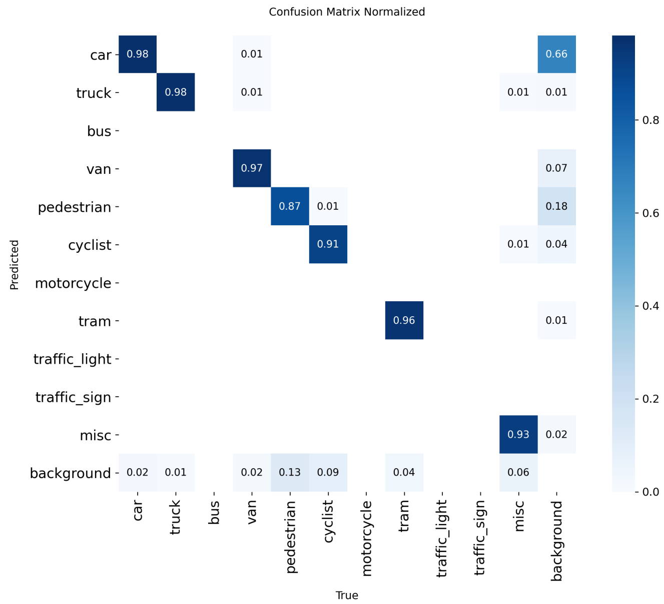

Figure 6.1: Normalised confusion matrix on the validation split. Diagonal entries are correct classifications; off-diagonal entries show the categories most often interchanged.

Table 6.2: Effect of training schedule and model capacity

Configuration	Model	Epochs	mAP@0.5	mAP@0.5:0.95
Baseline	YOLO11m	50	91.7	69.3
Extended	YOLO11m	500	94.2	77.7
Final	**YOLO11x**	**300**	**95.4**	**79.9**

ment of the depth model on the host processor rather than any property of the pipeline; on the accelerator the identical code is approximately twenty-five times faster. This is recorded because the two figures separate a deployable system from an unusable one while differing in no respect other than device placement — a diagnosis worth making before attributing latency to architecture.

6.4 Plausibility Filter

The filter removed 979 of 7,077 size-checked candidates. In an earlier configuration using a narrower tolerance under an uncalibrated focal length, the rejection rate reached approximately 27 % and included large numbers of vehicle classes, indicating suppression of genuine objects rather than spurious ones. Widening the tolerance in the uncalibrated regime, as described in Section 4.2, reduced the rate to 13.8 %.

The episode illustrates the intended asymmetry: an uncertain calibration should relax the admission criterion, not tighten it.

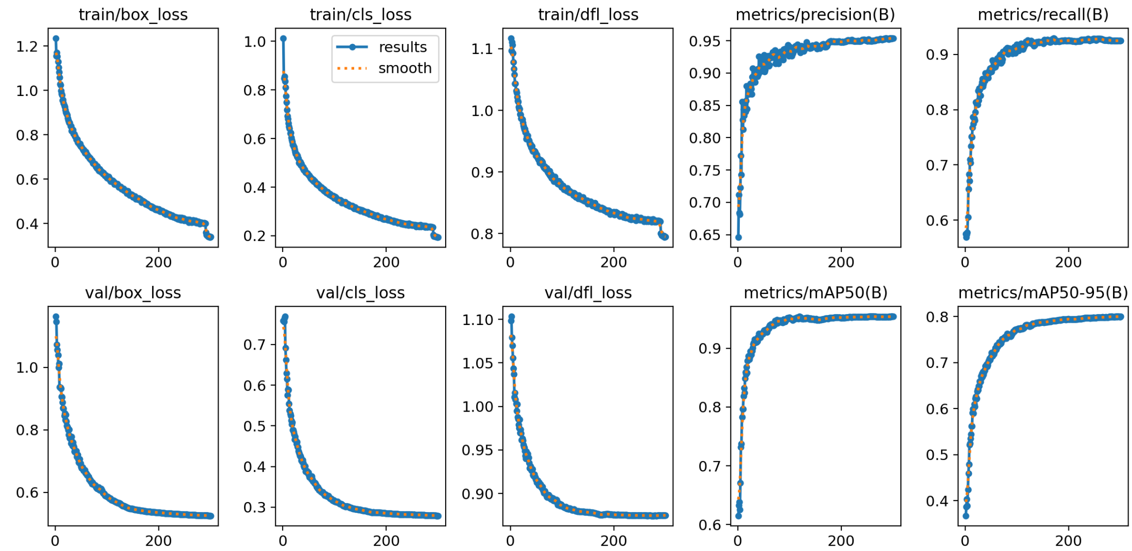

Figure 6.2: Training behaviour of the final configuration. Box, classification, and distribution-focal losses converge without divergence on either split, and both mAP measures rise monotonically before flattening.

Table 6.3: Complete pipeline over a 3,604-frame sequence

Measurement	Value
Frames processed	3,604
Throughput	23.1 fps (43 ms mean)
Collision alerts	69 (20 brake, 49 warning)
Frames with lane geometry	3,603 of 3,604
Detections rejected by filter	979 of 7,077 (13.8 %)
Detector parameters	56.8 M (194.5 GFLOPs)

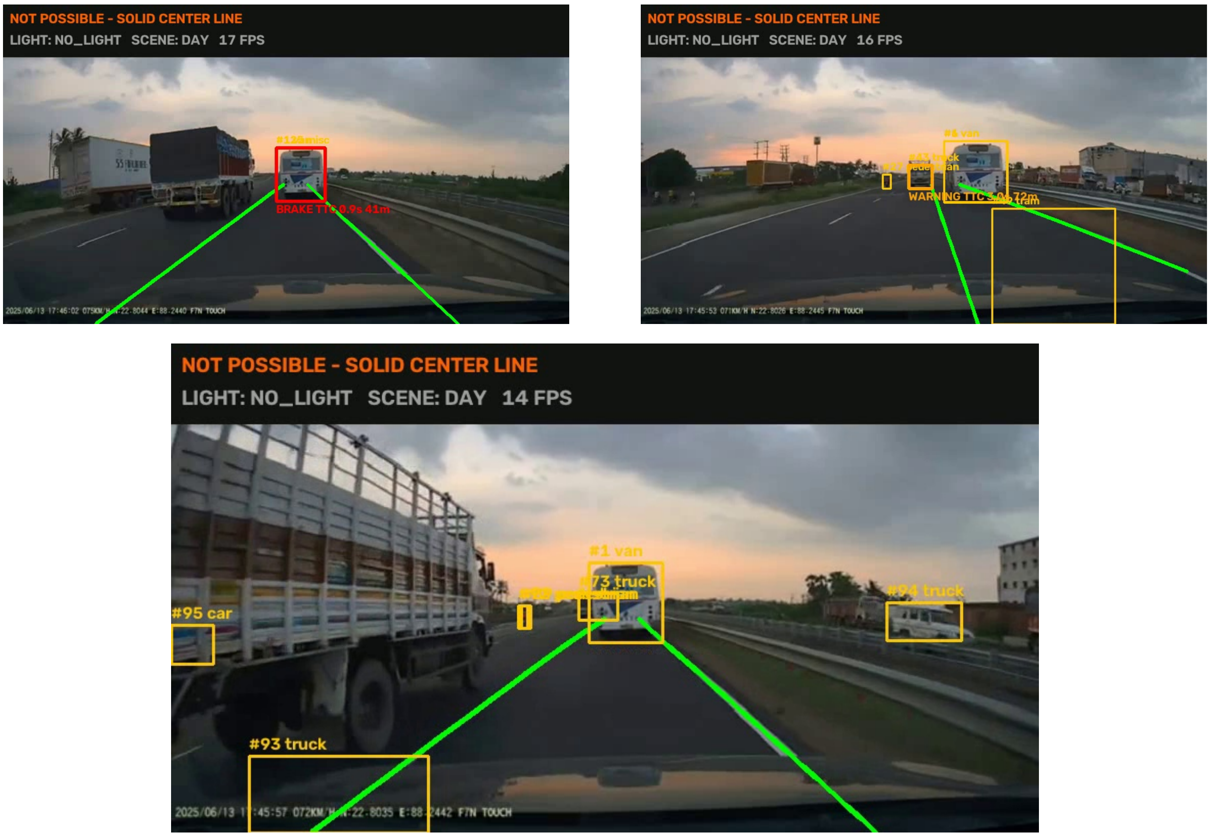

Figure 6.3: Pipeline output. Left: a brake-tier alert. Right: the warning tier. Below: simultaneous tracking under dense traffic. Every overlay was produced by the pipeline; none was added afterwards.

CHAPTER 7

Limitations

Each limitation below traces to a data deficiency rather than an implementation defect, and each therefore implies a different remedy.

7.1 Four Declared Classes Carry No Training Examples

The label space defines traffic lights, traffic signs, buses, and motorcycles, but the merge that produced this checkpoint drew from a single structured-road source containing none of them. Validation consequently reports seven classes rather than eleven. No modification to the model can address this; a corpus containing the categories is required.

7.2 India-Specific Categories Are Defined but Untrained

The autorickshaw, animal, rider, and fallback classes are present in the taxonomy, and conversion pipelines for three Indian corpora [3, 4, 5] are implemented and tested. The retraining that would populate these classes has not been performed. Reported accuracy therefore reflects the structured-road distribution and should not be read as performance on unstructured traffic.

7.3 The Overtaking Verdict Is Conservative Without Marking Type

Without a reliable solid-versus-broken classification, the legality condition cannot be satisfied and the verdict remains *not permitted*. This is the intended fail-safe rather than a malfunction, but it does mean the module is inactive on sequences lacking marking-type input.

7.4 Monocular Depth Is Less Reliable Than Active Ranging

Distance, and hence time-to-collision, inherits the error characteristics of learned monocular depth. A system with an active range sensor would obtain more dependable distances; the camera-only choice reflects a deployability constraint and should be recognised as a trade-off.

CHAPTER 8

Conclusion and Future Work

8.1 Conclusion

This report has presented a monocular perception pipeline integrating detection, tracking, lane and corridor estimation, metric depth, and two decision layers within a single pass, attaining 95.4 % mAP@0.5 and sustaining 23.1 frames per second on an H100 accelerator.

The system's distinguishing commitments are a degradation contract exercised under real deployment failure, and the expression of safety-relevant quantities in physical rather than image units. Its limitations are stated with their diagnosed causes rather than omitted.

8.2 Future Work

1. **Retraining on the merged Indian corpora.** The conversion infrastructure is complete. The improvement reported in [5] for comparable adaptation suggests the achievable magnitude. This is the highest-value next step.
2. **Incorporating a corpus containing traffic lights and signs**, rendering four presently untrainable classes trainable.
3. **Camera calibration**, which would narrow the plausibility filter from its uncertainty-widened interval and supply metric curvature.
4. **Supervised training of the drivable-corridor segmenter** to replace the present classical-vision substitute, improving lane estimation precisely where painted markings are unavailable.

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